Lucio La Cava, Ph.D.

Assistant Professor · DIMES Dept., University of Calabria

DIMES Dept., Via P. Bucci 44Z, University of Calabria, Rende (CS), Italy

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RESEARCH INTERESTS *Natural Language Processing* · *Machine-Generated Text Detection* · *LLM Interpretability* · *Representation Engineering* · *Argument Mining* · *Computational Social Science*

APPOINTMENTS

Assistant Professor (RTD-A) — Rende, Italy
DIMES Dept., University of Calabria — *May 2024 – present*

RESEARCH VISITS

Visiting Researcher — Copenhagen, Denmark
NeRDS Group, IT University of Copenhagen — *Aug. 2025 – Sep. 2025*

- Advisor: Prof. Luca Maria Aiello
- Project: COCOONS (COllective COordination through ONline Social media)

Visiting Ph.D. Student — Copenhagen, Denmark
NeRDS Group, IT University of Copenhagen — *Jan. 2023 – Apr. 2023*

- Advisor: Prof. Luca Maria Aiello
- Project: COCOONS (COllective COordination through ONline Social media)

FELLOWSHIPS & RESEARCH PROJECTS

Junior Fellow — Complex Human Behavior Lab — Trento, Italy
Fondazione Bruno Kessler (FBK) — *Jan. 2024 – present*

- Collaborating on AI-CODE (AI services for COntinuous trust in emerging Digital Environments), funded by the European Union

Team Member — PRIN 2022 Project — Rende, Italy
AWESOME: Analysis Framework for Web3 Social Media, DIMES Dept., Univ. Calabria — *Jun. 2023 – present*

- Approved for funding by MUR under the programme PRIN 2022, Line PE6 (Computer Science and Informatics)

EDUCATION

Ph.D. in Information and Communication Technologies — Rende, Italy
DIMES Dept., University of Calabria — *Nov. 2020 – Feb. 2024*

- Thesis: *"Graph Mining and Multimodal Representation Learning for Decentralized Socio-Economic Domains"*
- Supervisors: Prof. Sergio Greco and Prof. Andrea Tagarelli
- Final mark: **Excellent Cum Laude**

EDUCATION *continued*

M.Sc. in Computer Engineering — Rende, Italy
DIMES Dept., University of Calabria — *Sep. 2017 – Jul. 2020*

- Thesis: *"Decentralization of social networks in federation of instances: exploratory and generative profiling models in Mastodon"*
- Supervisor: Prof. Andrea Tagarelli · Final mark: **Magna Cum Laude**

HONORS & AWARDS

2026 — **National Scientific Qualification for Associate Professorship (ASN)** — *Italian MUR*
Scientific Disciplinary Area IINF-05/A — Information Processing Systems

2024 — **WebSci'24 Best Short Paper Award**
"Polarization in Decentralized Online Social Networks", sponsored by ACM SIGWEB

2023 — **AEGEAN Scholarship** — *Thessaloniki, Greece*
Top students at the Mediterranean Machine Learning (M2L) Summer School

2023 — **Zhipu Student Travel Award** — *Austin, Texas*
The ACM Web Conference 2023

2022 — **SIGIR 2022 Student Travel Grant** — *Madrid, Spain*
The 45th International ACM SIGIR Conference on Research and Development in Information Retrieval

2020 — **Italian qualification to practice as an Engineer** — *Univ. of Calabria*
Section A — Field: Information Technology (no exp. date)

PUBLICATIONS

JOURNAL ARTICLES 12

[J12] **La Cava, L.**, Aiello, L.M. & Tagarelli, A. *Machines in the crowd? Measuring the footprint of machine-generated text on Reddit.* **Online Social Networks and Media** 55, 100355 (2026). Q2 IF 2.9

[J11] Alfano, G., Greco, S., **La Cava, L.**, Mahmood, T. & Trubitsyna, I. *Semantic-aware data imputation in dynamic relational databases via pre-trained language models.* **Information Fusion** 2026. doi:10.1016/j.inffus.2026.104291 Q1 IF 15.5

[J10] **La Cava, L.**, Mandaglio, D., Zangari, L. & Tagarelli, A. *Heuristic-informed mixture of experts for link prediction in multilayer networks.* **Information Sciences** 2026. doi:10.1016/j.ins.2026.123106 Q1 IF 6.8

[J9] **La Cava, L.** & Tagarelli, A. *Safeguarding Decentralized Social Media: LLM Agents for Automating Community Rule Compliance.* **Online Social Networks and Media** 48, 100319 (2025). doi:10.1016/j.osnem.2025.100319 Q2 IF 2.9

[J8] Galdeman, A., **La Cava, L.**, Zignani, M., Tagarelli, A. & Gaito, S. *Characterizing NFT Markets through a Multilayer Network Approach.* **Blockchain: Research and Applications** 2025. doi:10.1016/j.bcra.2025.100356 Q1 IF 5.6

[J7] Antelmi, A., **La Cava, L.** & Pera, A. *Characterizing swing voters in online social media: the case of the 2022 Italian elections.* **Soc. Netw. Anal. Min.** 15, 77 (2025). doi:10.1007/s13278-025-01457-4 Q2 IF 2.8

[J6] Martirano, L., **La Cava, L.** & Tagarelli, A. *Unveiling User Dynamics in the Evolving Social Debate on Climate Crisis During the Conferences of the Parties.* **Pervasive and Mobile Computing** (2025). doi:10.1016/j.pmcj.2025.102077 Q2 IF 3.5

[J5] **La Cava, L.**, Costa, D. & Tagarelli, A. *Visually Wired NFTs: Exploring the Role of Inspiration in Non-Fungible Tokens.* **ACM Transactions on the Web** (2025). doi:10.1145/3703411 Q1

PUBLICATIONS *continued*

[J4] **La Cava, L.**, Aiello, L.M. & Tagarelli, A. *Drivers of social influence in the Twitter migration to Mastodon.* **Nature Scientific Reports** 13, 21626 (2023). doi:10.1038/s41598-023-48200-7 Q1 IF 4.6

[J3] Costa, D., **La Cava, L.** & Tagarelli, A. *Unraveling the NFT economy: A comprehensive collection of Non-Fungible Token transactions and metadata.* **Data in Brief** 51, 109749 (2023). doi:10.1016/j.dib.2023.109749 Q2 IF 1.2

[J2] **La Cava, L.**, Greco, S. & Tagarelli, A. *Information consumption and boundary spanning in Decentralized Online Social Networks: The case of Mastodon users.* **Online Social Networks and Media** 30, 100220 (2022). doi:10.1016/j.osnem.2022.100220 Q1 IF 2.9

[J1] **La Cava, L.**, Greco, S. & Tagarelli, A. *Understanding the growth of the Fediverse through the lens of Mastodon.* **Applied Network Science** 6, 64 (2021). doi:10.1007/s41109-021-00392-5 Q1 IF 2.2

CONFERENCE PROCEEDINGS 20

[C20] **La Cava, L.**, Monea, S. F. & Greco, S. *Who Argues What? Joint Argument-Entity Detection and Classification in Political Debates.* **EMNLP 2026 (Findings)**.

[C19] **La Cava, L.** & Tagarelli, A. *Toward Preference-aligned Large Language Models via Residual-based Model Steering.* **IJCAI 2026**. doi:10.24963/ijcai.2026/642 CORE A*

[C18] **La Cava, L.**, Macko, D., Moro, R., Srba, I. & Tagarelli, A. *Authorship Attribution in Multilingual Machine-Generated Texts.* **ACL 2026 (Main)**. doi:10.18653/v1/2026.acl-long.2091 CORE A* AR 19%

[C17] Caputo, E., **La Cava, L.** & Greco, S. *Argument Component Segmentation with Fine-Tuned Large Language Models.* **EACL 2026 (Findings)**. doi:10.18653/v1/2026.findings-eacl.272

[C16] Caputo, E., **La Cava, L.** & Greco, S. *ARGUS: Towards End-to-End Argument Mining with Large Language Models.* **AAAI 2026 (Demonstration Track)**. doi:10.1609/aaai.v40i48.42335

[C15] **La Cava, L.** & Tagarelli, A. *OpenTuringBench: An Open-Model-based Benchmark and Framework for Machine-Generated Text Detection and Attribution.* **EMNLP 2025 (Main)**. doi:10.18653/v1/2025.emnlp-main.1354 CORE A*

[C14] Greco, C. M., **La Cava, L.**, Zangari, L. & Tagarelli, A. *Exploring LLMs' Ability to Spontaneously and Conditionally Modify Moral Expressions through Text Manipulation.* **ACL 2025 (Main)**. doi:10.18653/v1/2025.acl-long.883 CORE A*

[C13] **La Cava, L.** & Tagarelli, A. *Open Models, Closed Minds? On Agents Capabilities in Mimicking Human Personalities through Open Large Language Models.* **AAAI 2025**. doi:10.1609/aaai.v39i2.32125 CORE A* AR 23.4%

[C12] Gugliotta, C., **La Cava, L.** & Tagarelli, A. *PersonaGen: A Persona-Driven Open-Ended Machine-Generated Text Dataset.* **CIKM 2025 (Resource Track)**.

[C11] Greco, C. M., **La Cava, L.** & Tagarelli, A. *Talking the Talk Does Not Entail Walking the Walk: On the Limits of Large Language Models in Lexical Entailment Recognition.* **EMNLP 2024 (Findings)**. doi:10.18653/v1/2024.findings-emnlp.881

[C10] **La Cava, L.**, Costa, D. & Tagarelli, A. *Is Contrasting All You Need? Contrastive Learning for the Detection and Attribution of AI-generated Text.* **ECAI '24**. doi:10.3233/FAIA240862 CORE A AR 23%

[C9] **La Cava, L.**, Mandaglio, D. & Tagarelli, A. *Polarization in Decentralized Online Social Networks.* **WebSci '24**. doi:10.1145/3614419.3644013

[C8] Bono, C., **La Cava, L.**, Luceri, L. & Pierri, F. *An Exploration of Decentralized Moderation on Mastodon.* **WebSci '24**. doi:10.1145/3614419.3644016

[C7] **La Cava, L.**, Costa, D. & Tagarelli, A. *SONAR: Web-based Tool for Multimodal Exploration of Non-Fungible Token Inspiration Networks.* **SIGIR '23**. doi:10.1145/3539618.3591821 CORE A*

[C6] Costa, D., **La Cava, L.** & Tagarelli, A. *Show me your NFT and I tell you how it will perform: Multimodal representation learning for NFT selling price prediction.* **WWW '23**. doi:10.1145/3543507.3583520 CORE A* AR 19.2%

PUBLICATIONS *continued*

[C5] Martirano, L., **La Cava, L.** & Tagarelli, A. *Evolution of the Social Debate on Climate Crisis: Insights from Twitter During the Conferences of the Parties.* **ICT-DM '23**. doi:10.1109/ICT-DM58371.2023.10286927

[C4] Gullo, F., **La Cava, L.**, Mandaglio, D. & Tagarelli, A. *When Correlation Clustering Meets Fairness Constraints.* **DS '22**. doi:10.1007/978-3-031-18840-4_22 CORE B

[C3] **La Cava, L.**, Simeri, A. & Tagarelli, A. *LawNet-Viz: A Web-based System to Visually Explore Networks of Law Article References.* **SIGIR '22**. doi:10.1145/3477495.3531668 CORE A*

[C2] **La Cava, L.**, Greco, S. & Tagarelli, A. *Network Analysis of the Information Consumption-Production Dichotomy in Mastodon User Behaviors.* **ICWSM '22**. doi:10.1609/icwsm.v16i1.19391 CORE A

[C1] **La Cava, L.**, Greco, S. & Tagarelli, A. *Discovering the Landscape of Decentralized Online Social Networks through Mastodon.* **SEBD '22**.

WORKSHOP PAPERS 4

[W4] Gullo, F., **La Cava, L.**, Mandaglio, D. & Tagarelli, A. *Correlation Clustering with Fairness Constraints.* **Green-Aware AI @ ECAI 2025**.

[W3] **La Cava, L.**, Mandaglio, D. & Tagarelli, A. *Polarized Communities in Mastodon: Insights from Instance-Level Analysis.* **SAI4OID @ WI-IAT 2025**.

[W2] Galdeman, A., **La Cava, L.**, Zignani, M., Tagarelli, A. & Gaito, S. *Clubs of wallets: a dive into the mesoscopic features of NFT transaction networks.* **ComplexDataBlocks @ CCS 2022**.

[W1] **La Cava, L.**, Simeri, A. & Tagarelli, A. *The Italian Civil Code Network Analysis.* **RELATED @ ICAIL 2021**.

INVITED TALKS & TUTORIALS

INVITED TALKS

6 Feb. 2025 **Researching Next-Gen Social Media Ecosystems & Scientific Research in the Age of Big Players**
NetPLACE Seminars, Online

30 Jan. 2024 **Discovering Decentralized Online Social Media through the lens of Mastodon**
Invited talk · Oxford Internet Institute, University of Oxford, UK

TUTORIALS

2 Sep. 2024 **Mining, Modeling, and Analyzing Decentralized Social Media (with A. Galdeman)**
ASONAM 2024, Univ. Calabria, Rende, Italy

6 Nov. 2023 **The Rise of the Fediverse: Shaping the Future of Social Media Research through Mastodon**
ASONAM 2023, Kusadasi, Turkey

TEACHING

LECTURER · UNIVERSITY OF CALABRIA

2026 – present **Machine Learning** · B.Sc. in Computer Engineering 38h

2025 – 2026 **Data Mining and Bioimages** · M.Sc. in Medicine and Surgery — Digital Technologies 39h

2024 – 2025 **Algorithms and Data Structures** · B.Sc. in Computer Engineering (Bioengineering) 19h

INVITED LECTURER

May 2025 **Large Language Models: from Theory to Applications** 10h
PhD-level Course · Uppsala Univ., Sweden

TEACHING *continued*

TEACHING ASSISTANT · UNIVERSITY OF CALABRIA

2026 – present	**Machine Learning** · B.Sc. in Computer Engineering	18h
2025 – present	**Data Mining** · B.Sc. in Computer Engineering (AI & ML)	18h
2024 – 2025	**Algorithms and Data Structures** · B.Sc. in Computer Engineering (Bioengineering)	9h
2021 – 2024	**Data Mining** · M.Sc. in Computer Engineering (AI & ML)	16h
2023 – 2024	**Algorithms and Data Structures** · B.Sc. in Computer Engineering	18h
2022 – 2023	**Social Networks and Media Analysis** · M.Sc. in Computer Engineering	8h
2021 – 2023	**Big Data Analytics** · M.Sc. in Computer Engineering for the IoT	16h
2020 – 2021	**Networks and Systems in Bioinformatics** · M.Sc. in Computer Engineering (Big Data & Bioinformatics)	16h

THESIS SUPERVISION

2024 – 2025	*Argument Mining and Large Language Models: Identification and Classification of Argument Components*	*M.Sc. thesis*
2024 – 2025	*Detection of Texts by Large Language Models Conditioned on Individual Profiles*	*M.Sc. thesis*
2024 – 2025	*Design and Development of a System for the Analysis of Bank Frauds*	*B.Sc. thesis*
2024 – 2025	*Phishing and Artificial Intelligence: Perspectives for Red Teams and Blue Teams*	*B.Sc. thesis*
2023 – 2024	*Modeling and structural analysis of voting relationships in blockchain-based online social media: Steemit as a case study*	*M.Sc. thesis*
2022 – 2023	*User Recommendation in Decentralized Online Social Networks*	*M.Sc. thesis*
2022 – 2023	*Non-Fungible Tokens: technology, security and applications*	*B.Sc. thesis*
2022 – 2023	*Personal ChatBot: generative artificial intelligence for the understanding of individual personality*	*M.Sc. thesis*
2022 – 2023	*Pre-trained neural language models for classification and text generation of social media content creators*	*M.Sc. thesis*
2022 – 2023	*Topic modeling and affective computing in the languages of different social media content creator roles*	*M.Sc. thesis*
2022 – 2023	*Computer vision and digital art: NFT image semantic search based on visual neural models and generation of textual descriptions*	*M.Sc. thesis*

PROFESSIONAL SERVICE

EDITORIAL ROLES

2026 – present	**Social Network Analysis and Mining, Springer**	*Associate Editor*
2025 – present	**Computational Intelligence, Wiley**	*Associate Editor*
2023	**Special Issue on Decentralized Online Social Networks, OSNEM, Elsevier**	*Guest Editor*

ORGANIZATION

2025	**PersonaLLM: Workshop on LLM Persona Modeling @ NeurIPS 2025**	*Co-organizer*
2025	**Next-Gen and Alternative Social Media 2025 @ AAAI ICWSM**	*General Co-Chair*

PROFESSIONAL SERVICE *continued*

2024	**DeWeb 2024: 1st Int. Workshop on Decentralizing the Web @ AAAI ICWSM**	*General Co-Chair*
2024	**ASONAM 2024: 16th Int. Conf. on Advances in Social Networks Analysis and Mining**	*Workshops Chair, Session Chair*

PROGRAM COMMITTEES & REVIEWING

Conferences	AAAI, ICLR, IJCAI, ECAI, TheWebConf
Workshops	MIDAS @ ECML-PKDD, LIMBO @ ECML-PKDD